Lecture 1: Introduction to Innovation

IT4050 Innovation Management and Entrepreneurship

- Invention vs innovation
- Why innovation matters
- Types of innovation
- Drucker's 7 sources

Invention vs Innovation

Not every invention is an innovation

	Invention	Innovation
Meaning	Something new that never existed before	Something new that adds **value to customers**
Needs customers?	No	Yes
Patent idea	Inventive step (non-obviousness)	Knowledge store grows

Example: Xerox PARC → Apple

Invention became innovation when it reached customers

- **Xerox PARC** invented the mouse and GUI (invention)
- **Apple** developed and sold them to the world (innovation)
- Also: microprocessor, telephone, printing press, steam engine, Internet

Why innovation matters

Innovation management = using opportunities to introduce new ideas, processes or products

- **Survival**: sustains the business long term
- **Customers**: higher satisfaction
- **Competition**: advantage and differentiation
- New markets, better quality, growth in profit

Types: product, service, process

Product

a better smartphone

Service

ride hailing, internet banking

Process

Ford's automated assembly line

Incremental vs Radical

	Incremental	Radical
Change	Small improvements	Major breakthrough
Risk	Low	High
Result	Keeps current customers	New markets, old products obsolete
Example	Phone with a better camera	First smartphone, Uber

Architectural vs Modular

Trick: architectural = links, modular = a part

Architectural

links between parts change

Sony Walkman, smartphone

Modular

one part changes

hard disk to SSD

Peter Drucker's 7 sources of innovation

Learn this list word for word

- 1 The unexpected
- 2 Incongruity
- 3 Process need
- 4 Changes in industry / market structure
- 5 Demographics
- 6 Changes in perception, mood and meaning
- 7 New knowledge

Drucker's 7: examples

Source	Example
The unexpected	Penicillin (Fleming)
Process need	Ford assembly line
Demographics	Ageing population → health apps
Perception change	Health awareness → organic food
New knowledge	AI, nanotechnology
Market structure	Uber Eats

Recap: Lecture 1

- Invention = new; innovation = new + value to customers
- Types: product, service, process
- Incremental vs radical; architectural vs modular
- Drucker's 7 sources

Lecture 2: Innovation Management Techniques

Tools that make innovation systematic

- The 9 IMTs
- Knowledge management and IPR
- Brainstorming, mind mapping, prototyping
- QFD and business simulation

The 9 IMTs

Memorise the list

1 Knowledge management

2 Intellectual property rights

3 Customer relationship management

4 Business intelligence

5 Brainstorming

6 Mind mapping

7 Prototyping

8 Quality function deployment

9 Business simulation

Knowledge management and IPR

- **KM**: creating, sharing, using and managing knowledge
- Dimensions: **tacit** (in people's heads) and **explicit** (documented)
- **IPR** strategic use: **Sell · License · Collaborate · Donate**

CRM and Business intelligence

- **CRM**: gather, manage and use customer information
- Interviews about pain points drive product development
- **BI**: handle large structured and unstructured data to find strategic opportunities

Brainstorming

Followed by affinity grouping

- Go for **quantity**
- **Withhold criticism**
- Welcome **wild ideas**
- **Combine and improve** ideas

Mind mapping and Prototyping

Mind map

non-linear diagram around a central concept

Prototype

an early sample or model built to test a concept

Rapid prototyping: 3D CAD scale model

QFD and Business simulation

- **QFD**: qualitative user demands → quantitative parameters (main tool: House of Quality)
- **Business simulation**: test decisions without real risk (process, marketing simulation games)

Recap: Lecture 2

- 9 IMTs
- KM: tacit vs explicit
- IPR: Sell, License, Collaborate, Donate
- Brainstorming: 4 rules + affinity grouping

Lecture 3: Ideation and Creativity

Where innovation starts: ideas

- Ideas and creativity
- Wallas's creative process
- Idea generation techniques
- Opportunity

Ideation and Creativity

- **Ideation**: generating, developing and communicating new ideas
- **Creativity**: the ability to generate new, original ideas and solutions
- Creativity is the building block of innovation

Wallas's creative process (1926)

Order matters: P I I V

#	Stage	Meaning
1	Preparation	gather knowledge, understand the problem
2	Incubation	the subconscious works on it
3	Illumination	the sudden "aha" moment
4	Verification	evaluate and test the idea

Idea generation techniques

- Role playing
- Active search
- Attribute listing
- Brainstorm
- Critique

- Collaborate
- Documenting
- Expert opinion
- Empathy / user research
- Storyboards

Inbound vs Outbound ideation

Inbound

internal ideas

brainstorming, employee ideas, rewards

Outbound

voice of customer (focus groups, lead users)

open innovation (idea contests)

miscellaneous (patent mapping)

Enhancing creativity

- Creativity training
- Support, time and tools
- Reward creativity
- Model creative behaviour
- Expect creativity and tolerate failure
- Encourage curiosity ("what if...")

Barriers to creativity

- Searching for the one "right" answer
- Following rules blindly, always being practical
- Becoming overly specialised
- Fearing mistakes and failure
- Believing "I'm not creative"

From idea to opportunity

Opportunity = a favourable set of circumstances that creates a need for a new product, service or business

- 1 Strategic fit
- 2 Business plan
- 3 Team
- 4 Leadership
- 5 Resources

Recap: Lecture 3

- Wallas: Preparation, Incubation, Illumination, Verification
- 10 techniques; inbound vs outbound
- How to enhance creativity and the barriers
- Idea to opportunity: 5 elements

Lecture 4: New Product Development

From idea to a product customers can buy

- NPD: 8 stages
- Stage-Gate
- Design thinking
- House of Quality
- Innovation sweet spot

NPD: the 8 stages

New product development = bringing a new product to market

1 Idea generation

2 Idea screening

3 Concept development and testing

4 Market strategy and business analysis (4 Ps)

5 Feasibility study

6 Technical design and roadmap

7 Test marketing

8 Market entry and commercialization

NPD example: Uber

- 2008: idea (no taxi on a snowy night in Paris)
- App built with Python, Node.js, Java
- 2010: first ride in San Francisco
- Dec 2011: launch in Paris, then worldwide

Stage-Gate process

Stages collect information; gates decide

- **Stage**: collect information and do the work
- **Gate**: decision point: Go / Kill / Hold / Recycle
- Each gate has criteria to fund the next stage
- Success drivers: customer focus, tough Go/Kill, cross-functional teams, a proven process

Design thinking: 5 steps

Human-centred and iterative

- 1 **Empathize**: understand the user
- 2 **Define**: the real problem
- 3 **Ideate**: brainstorm solutions
- 4 **Prototype**: weed out unworkable ideas
- 5 **Test**: get feedback

Design thinking vs Stage-Gate

	Design thinking	Stage-Gate
Focus	Empathy, the right problem	Managing risk and investment
Flow	Iterative	Linear stages with gates
Decisions	Learn from prototypes	Go / Kill by management
Best for	Early, uncertain stages	Funded development and launch

House of Quality (QFD)

Originated in 1972 at Mitsubishi's Kobe shipyard

- 1 Customer attributes (voice of the customer)
- 2 Weight their importance, compare with competitors
- 3 Engineering characteristics (measurable)
- 4 Relationship matrix
- 5 **Roof matrix**: trade-offs
- 6 Target values

Innovation sweet spot (IDEO)

A successful innovation sits where three circles overlap

Circle	Question
Desirability	Are we solving the right customer pain point?
Feasibility	Can we build and deliver it?
Viability	Can we make a profit and sustain it?

Recap: Lecture 4

- NPD: 8 stages in order
- Stage-Gate: stage + gate (Go/Kill/Hold/Recycle)
- Design thinking: E D I P T
- House of Quality: CA → EC → matrix → roof → targets
- Sweet spot: Desirability + Feasibility + Viability

Lecture 5: Entrepreneur and Entrepreneurship

Who is an entrepreneur?

- Stevenson's definition
- 4 risks
- Skills and growth mindset
- Danhof's types
- Intrapreneurship and social entrepreneurship

Who is an entrepreneur?

Howard Stevenson (Harvard)

- Entrepreneurship is the pursuit of opportunity **beyond resources controlled**
- An entrepreneur perceives an opportunity and creates an organisation to pursue it
- Founders often **bootstrap** (use their own funds)

The 4 risks of an entrepreneur

Risk	Question	Reduce it by
Demand	Will customers adopt it?	Interviews, MVP, pre-orders
Technology	Is a breakthrough needed?	Prototypes, partners
Execution	Can we attract the right team?	Strong team, advisers
Financing	Will capital be available?	Bootstrapping, staged funding

Traits, skills and growth mindset

Growth mindset: abilities can be learned and improved (fixed mindset: they cannot)

Traits

action oriented, motivated

calculated risk taker

determined, self-confident

creative, goal setter

Skills

financial skills · networking

accepting feedback · pattern recognition

strategic thinking · negotiation

growth mindset

Wage vs Entrepreneurial employment

	Wage employment	Entrepreneurial employment
Income	Fixed salary	Profit, uncertain
Risk	Low (employer carries)	High (owner carries)
Decisions	Follow the employer	Own strategy
Wealth creation	Mostly earn and spend	Creates jobs, products, national wealth

Danhof's 4 types of entrepreneurs

Type	Idea	Example
Innovative	introduces new products or methods	Dilmah (Merrill J. Fernando)
Imitative (Adoptive)	copies and adapts proven innovations	PickMe, Softlogic
Fabian	timid; changes only if forced	traditional family shop
Drone	refuses to change	-

Intrapreneur vs Entrepreneur

Social entrepreneurship: business methods to solve social problems; social impact is the main goal

	Entrepreneur	Intrapreneur
Business	Own	Inside an existing organisation
Capital and risk	Personal	The organisation's
Reward	Profit, independence	Salary, bonus, promotion

Networking and funding

- Networking: mentors, customers, partners, talent, funding
- Funding: savings, family, angels, venture capital, bank loans, crowdfunding, grants, incubators
- To attract funding: clear plan and pitch, traction, strong team, realistic financials, an exit

Recap: Lecture 5

- Stevenson: opportunity beyond resources controlled
- Risks: Demand, Technology, Execution, Financing
- Danhof: Innovative, Imitative, Fabian, Drone
- Intrapreneur works inside an organisation

Lecture 6: Value Proposition and Business Model

35 marks in the 2025 sample paper

- Value Proposition Canvas
- Business Model Canvas (9 blocks)
- Lean Canvas
- Customer Factory
- How to answer canvas questions

Value Proposition Canvas (VPC)

Fit = pain relievers and gain creators match the customer's important pains and gains

Customer profile

Jobs: what they try to get done

Pains: bad outcomes, risks

Gains: benefits they want

Value map

Products and services

Pain relievers

Gain creators

Validate your idea

- **Inventor's bias**: loving your solution without evidence
- Love the problem, start with the customer profile
- Validate with interviews, surveys, MVP, pre-orders, A/B tests
- Google Glass: a solution looking for a problem

Business Model Canvas: 9 blocks

Left = product side, right = market side, middle = value proposition

Key partners

Key activities

Key resources

Cost structure

Value propositions

Customer segments

Channels

Customer relationships

Revenue streams

BMC: the key question of each block

Block	Ask
Customer segments	For whom do we create value?
Value propositions	What problems do we solve?
Channels, relationships	How do we reach and keep them?
Revenue streams	What do they pay for?
Resources, activities, partners	What do we need, do and who helps?
Cost structure	What are the biggest costs?

Lean Canvas vs BMC

- **9 blocks**: Customer segments · Problem · Unique value proposition · Solution · Channels · Revenue streams · Cost structure · Key metrics · Unfair advantage
- Replaces partners, activities, resources, relationships with problem, solution, key metrics, unfair advantage
- Better for start-ups: starts with the problem, measures traction, has an unfair advantage

Customer Factory model

Goal: make happy customers

- 1 **Acquisition**: attract visitors
- 2 **Activation**: first good experience
- 3 **Retention**: they come back
- 4 **Revenue**: they pay
- 5 **Referral**: they bring others

How to answer a canvas question

- Read the case once and underline the facts
- Draw the grid; 2 to 4 short bullets per block
- **Fill every block**; label assumptions
- Two-sided platform: split the segments (guests / hosts)
- VPC: profile first, then value map

Recap: Lecture 6

- VPC: jobs, pains, gains | products, pain relievers, gain creators
- BMC: 9 blocks
- Lean Canvas: problem, solution, key metrics, unfair advantage
- Customer Factory: A A R R R

Lecture 7: Business Plan and Investors

The 15-mark question lives here

- Why a business plan
- Mission, vision, SMART objectives
- Executive summary
- The 3 tests
- Pitching to investors

Why a business plan? Main sections

- A **road map** for the business; for banks, investors and employees
- Value is in the thinking and research, not only the document
- Sections: executive summary · company description · products and services · marketing plan · operational plan · management · start-up expenses · financial plan · appendices

Mission, Vision, Values, Objectives

	Meaning
Mission	Why the business exists now: purpose, what it does
Vision	Where it will go in the future
Values	Triple bottom line: profit, people, planet
Objectives	**SMART**: Specific, Measurable, Achievable, Relevant, Time-bound

Executive summary

Written last, presented first

- About 2 pages, positive, use numbers
- Answers 5 questions:
- 1 Is there an attractive market?
- 2 Products or services to win?
- 3 People to make it happen?
- 4 How much money will it make?
- 5 Where will the business go?

The 3 tests every business plan must pass

Barringer and Ireland (check against your own notes)

Test	Question	Evidence
Reality	Is there a real market and can it be built?	Research, interviews, prototype
Competitive	Can we build a lasting advantage?	Competitor analysis, UVP
Value	Is it financially attractive?	Forecast, cash flow, ROI

Pitching to investors: 4 things to emphasise

#	Emphasise	Why the investor cares
1	The **team**	People deliver the plan
2	**Problem, solution, market**	Proves demand
3	**Traction and business model**	Evidence it works
4	**Financials, the ask, the exit**	Return and risk

What investors investigate

Bankers want repayment; investors want growth and a share of the rewards

- Management team and track record
- Market size and demand
- Product, technology and IP
- Competition and competitive advantage
- Business model and financial projections
- Legal and regulatory position
- Exit options

Recap: Lecture 7

- Business plan sections; executive summary: 5 questions
- SMART objectives; mission vs vision
- 3 tests: Reality, Competitive, Value
- Pitch: Team, Problem-solution-market, Traction, Financials-ask-exit

Lecture 8: Marketing and Operational Plan

How the business sells and runs every day

- 4 Ps and 4 Cs
- Marketing plan
- Channels and lead generation
- Operational plan

4 Ps and 4 Cs

Marketing = create, communicate and deliver value

4 Ps (seller)	4 Cs (customer)
Product	Customer solution
Price	Customer cost
Place	Convenience
Promotion	Communication

Marketing plan: what goes in

- Industry economics, market size, barriers to entry
- Product: features vs benefits
- Customers: demographics and **persona**
- **Competitive analysis**: compare with 3 main competitors
- Niche, promotion and budget, pricing, location
- Sales forecast (best and worst case)

Channels, lead generation and nurturing

Lead generation

content marketing and blogs

SEO

social media ads

webinars, free trials

referral programs, landing pages

Lead nurturing

email drip campaigns

personalised follow-up

retargeting ads

newsletters

first-purchase discount

Operational plan

- Production: methods, quality control, customer service, inventory
- Location: space, utilities, access, cost
- Legal environment: licences, permits, insurance, patents
- Personnel and suppliers
- Credit policies
- Management and organisation

Recap: Lecture 8

- 4 Ps ↔ 4 Cs
- Persona and competitive analysis
- Lead generation vs lead nurturing
- Operational plan elements

Lecture 9: Financial Planning and Cash

Profit is not cash

- Why financial planning
- P&L, balance sheet, cash flow
- Budget and proforma
- Cash budget

Why financial planning matters

- Shows how the business will afford its goals
- Funds in and out, cash available, projected position
- Keeps liquidity and shows objectives are achievable
- **3 components**: budgeting · proforma financial statements · cash management

The 3 financial statements

Accounting equation: Assets = Liabilities + Capital

Statement	What it shows
Profit and loss	Income and expenses over a period (profitability, not cash)
Balance sheet	Assets = Liabilities + Equity at one point in time
Cash flow statement	Cash in and out: operating, investing, financing

Profit is not cash

Say this in every cash flow answer

- Profit = revenue – expenses
- Cash = money free and readily available
- A profitable business can fail if customers pay late
- Most start-ups fail from running out of cash, not lack of ideas

Budget and proforma statements

- **Budget**: estimate of income and expenses for a future period; creates discipline
- **Proforma statements**: projected income statement, balance sheet, cash flow, based on assumptions
- Used in loan and investment proposals; weakness: assumptions may be wrong

Cash budget: 5 steps

A cash map: amount and timing of receipts and payments

- 1 Set a minimum cash balance
- 2 Forecast sales
- 3 Forecast cash receipts (credit terms)
- 4 Forecast cash payments
- 5 Estimate the month-end balance; plan borrowing or investing

Recap: Lecture 9

- Financial planning: budgeting, proforma, cash management
- P&L, balance sheet, cash flow statement
- Profit is not cash
- Cash budget: 5 steps